Danika Heaney

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Education

University of California, Berkeley (Berkeley, CA)
PhD Student College of Chemistry
(2024-Present)

- Developed spatially resolved CO_2 emission maps to evaluate emissions reductions, air quality, and climate goals for the SF Bay Area working with Prof. Ronald Cohen
- Coursework in Environmental Fluid Mechanics, Atmospheric Chemistry, and Data Science

Wellesley College (Wellesley, MA)
BA in Chemistry and Minor in Computer Science
(2019-2023)

- Courses: The Climate Crisis, Physical, Organic, Inorganic, and Analytical Chemistry
- Leadership: Varsity Women's Soccer Captain, Chemistry Society Secretary, Developing a More Inclusive Chemistry Community Working Group, Club Frisbee

Research & Work Experience

Graduate Researcher
UC Berkeley (August 2024 – expected May 2029)

- Developed multi-year CO_2 maps for the SF Bay Area and Los Angeles using Bayesian Inverse Modeling to evaluate progress toward regional climate goals and identify priority areas for emissions reductions
- Reconciled discrepancies between existing bottom-up inventories and real-world sensor observations, to improve and expand the accuracy of public emissions data
- Handled complex environmental datasets for the Cohen Lab, while supporting publicly accessible data on our website
- Synthesized data from transportation, heating, and industrial sectors to construct comprehensive sector-specific emission trends, identifying key decarbonization opportunities to inform regional climate policy and planning
- Managed daily operations for the $BEACO_2N$ network, a high-density sensor array. Lead calibration schedules and data quality assurance to ensure accurate real-time climate monitoring

Fulbright Researcher
Technical University of Munich (September 2023 – July 2024)

- Accelerated the discovery of sustainable anode and cathode materials for lithium-ion batteries by developing automated robotics and machine learning workflows, contributing to the next generation of green energy storage solutions
- Collaborated with a majority international research team under Professor Helge Stein to advance green energy technology
- Served as a Teaching Assistant for Programming and Data Science (CH0103), an intro-level computer science and chemistry course with 100+ students

Summer Research Fellow - Atmospheric Chemistry
Stony Brook University (May 2021 – August 2021)

- Studied nitrate and iodate aerosol clusters using density functional theory to understand the new particle formation processes in the atmosphere, contributing to climate change research under Professor Christopher Johnson
- Applied computational chemistry methods to environmental problems with direct implications for understanding radiative forcing in the climate system

Research Assistant
Wellesley College (January 2019 – March 2022)
- Developed expertise in laboratory research by reassembling and operating an Ultra High Vacuum chamber and helium cryostat to study astrochemistry and prebiotic molecular formation in space
- Collaborated with undergraduate and graduate research students across multiple projects, facilitating communication and work delegation
- Mentored first-year students interested in lab work and astrochemistry
- Unofficial "tour-guide" for the lab showing new students about our work

Summer Research Fellow - Ultrafast 2DIR Spectroscopy
Boston University (May 2022 – August 2022)
- Studied supercritical fluids using a Greenhouse Gas with pump-probe and 2D IR spectroscopy under Professor Lawrence Ziegler
- Conducted data analysis and spectral predictions using MATLAB

Publications

The Competition between Hydrogen, Halogen, and Covalent Bonding in Atmospherically Relevant Ammonium Iodate Clusters
- Nicoline C. Frederiks, Danika D. Heaney, John J. Kreinbihl, and Christopher J. Johnson. The Competition between Hydrogen, Halogen, and Covalent Bonding in Atmospherically Relevant Ammonium Iodate Clusters. *Journal of the American Chemical Society* 2023 145 (2), 1165-1175 DOI: 10.1021/jacs.2c10841

Presentations

Trends in California CO_2 Emissions from Transportation and Building Energy seen through Independent Atmospheric Observations and a Bayesian Inversion
American Geophysical Union Annual Meeting, San Francisco (2026)

Mapping Temporal and Sectoral Trends in Urban CO_2 Emissions
Graduate Research Conference, UC Berkeley (September 25, 2025)

Expanding CO_2 Monitoring Insights beyond the BEACO_2N Network
Summer School for Inverse Modeling of Greenhouse Gases (July 10, 2025)

Automated Robotic Pipeline for Sustainable Lithium Ion Battery Formation
Technical University of Munich (June 20, 2024)

Rotational Coherences in Supercritical Fluids Observed in Pump-Probe Femtosecond Spectroscopy
Boston University Chemistry Research Symposium (August 19, 2022)

Investigating Iodine Containing Structures Involved in Atmospheric New Particle Formation

Virtual Summer Research Program Symposium (August 6, 2021)

Chemistry of the Heavens: The Cradle of Life
Boston University Chemistry Research Symposium (August 19, 2022)

Applying Hard Sciences to Hard Problems
Ruhlman Conference (May 8, 2021)

Awards

A&WMA Sustainability Research and Study Related to Air Quality Scholarship
Air & Waste Management Association (2026)

NSF Graduate Research Fellowships Program Honorable Mention
National Science Foundation (2024)

Fulbright Research Award
German American Fulbright Commission (March 2023)

American Chemical Society Undergraduate Award in Physical Chemistry
Wellesley College (May 2023)

Sigma Xi Research Honor Society
Wellesley College (May 2023)

Jean Crawford Award for Academic Excellence in Chemistry
Wellesley College (May 2023)

Massachusetts Space Grant 2020 and 2021 Fellowship Recipient
Wellesley College (September 2020-January 2021)

Teaching & Mentoring

Graduate Student Instructor
UC Berkeley (August 2024 – May 2026)

- Taught General Chemistry Laboratory (CHEM 1AL) to over 100 students across multiple lab sessions, encouraged scientific curiosity and hands-on lab skills
- Received overwhelmingly positive teaching feedback, with 74% of students rating my teaching as extremely effective (the highest possible score) on end-of-semester course evaluations, exceeding department averages across all survey questions
- Worked with lead instructor Michelle Douskey to create inclusive learning environments and support students from diverse academic backgrounds, particularly students from underserved high schools without previous advanced chemistry classes
- Also taught advanced Atmospheric Chemistry and Physics Laboratory with Professor Kristie Boering

College of Chemistry Transfer Student Mentor
UC Berkeley (August 2025 – Current)

- Mentoring three first-gen community college transfer students with biweekly meetings through their transition to UC Berkeley's rigorous chemistry program
- Successfully supported one mentee in joining a competitive chemistry research laboratory, opening doors for his scientific career

Environmental Education Instructor
The Nueva School, San Mateo, CA (January 2026)

- Developed a 70-minute interactive lesson on CO_2 emissions and air quality for high school students
- Connected real-world local air quality data from our Bay Area research to student learning modules in Python, bridging academic research and public education

Chemistry Tutor
Wellesley College (August 2022–January 2023)

- Employed as a tutor for students in both inorganic and organic chemistry. My role involved conducting group workshops and also personalized tutoring sessions, with a particular focus on physical chemistry

Leadership & Outreach

Bay Area Scientists Inspiring Students (BASIS) - Steering Committee Member
Community Resources for Science (August 2024 – Current)

- Lead coordination of Cohen Lab's school visit program, bringing hands-on atmospheric science education to 14 elementary schools across the Bay Area, reaching over 500 students annually
- Taught "Clouds, Clouds, Everywhere" atmospheric science curriculum in underserved Oakland and Berkeley schools, ensuring students from communities disproportionately affected by poor air quality and climate impacts have access to environmental education
- Served as panelist for Berkeley Unified School District's Climate Change Education initiative, advising teachers on effective approaches to teaching climate science and environmental issues to young students
- Helped organize logistics for campus-wide annual BASIS meeting

Developing a More Inclusive Chemistry Community - Working Group Member
Wellesley College Chemistry Department (September 2021 – January 2022)

- Collaborated with faculty and other students to identify and address barriers to inclusion in the Chemistry Department
- Advocated for expanded range of laboratory coat sizes to accommodate diverse body types

Chemistry Society Secretary
Wellesley College Chemistry Society (January 2023 – March 2023)

- Organized society meetings, took notes, and planned community-building events for chemistry students

Varsity Soccer Captain
Wellesley College (August 2021 – March 2023)

- Led 30+ teammates as captain for two seasons, fostering team cohesion and supporting players' athletic and personal development both on and off the field
- Developed leadership skills in motivation, conflict resolution, and building inclusive team culture